SILVER OAK UNIVERSITY
EDUCATION TO INNOVATION

Minor Project
1010063491
A.L.E.R.T.

Automated Landmark-based Eye and Response Tracker

Project ID	**CE_AIML_31**
Enrollment No	**2301030600043, 2301030600045, 2301030600049**
Name	**Prem Deshani, Kavya Kothari, Mehulsinh Rathod**
Branch	**B.Tech. Computer Engineering (AI & ML)**
Guide Name	**Ms. Vishwa Patel**

Index

- Introduction and Problem Context
- Background, Motivation and Relevance
- Literature Review and Research Gap
- Objectives and Milestones Achieved
- Dataset, Tools and Hardware
- Methodology: Kinematic Channels and Ordinal Loss
- Architecture: Edge Model Triad and Ensemble
- Results: Holdout Performance and Edge Latency
- System Verification, Conclusion and Future Work
- References

Introduction

- **Road Safety Crisis**: Driver fatigue and micro-sleeps contribute to over 20% of fatal collisions worldwide; road crashes cause 1.19 million deaths a year [1], [2].
- **Regulatory Mandates**: EU GSR 2019/2144 and Euro NCAP 2026 require real-time Driver Monitoring Systems (DMS) in new passenger vehicles [15].
- **ALERT Edge Solution**: A fully offline framework that extracts 468 3D facial landmarks with MediaPipe Tasks to track driver vigilance [3], [5], [14].
- **Temporal Deep Learning**: A heterogeneous ensemble of CNN-GRU, TCN and BiGRU networks reads 12 kinematic channels over 8-second windows.
- **Commodity Hardware**: A single CNN-GRU model runs at 65.3 FPS on a standard CPU within a footprint under 800 MB, with no in-cabin GPU required.

*Proprietary material of SILVER OAK UNIVERSITY

Background and Motivation

- **Physics of Micro-Sleeps**: A 2.5 s micro-sleep at 100 km/h means about 70 m of uncontrolled travel [1], [4].
- **Cloud DMS Unreliability**: Cloud systems add 150–500 ms of latency jitter, incur cellular costs and fail in tunnels and rural dead zones.
- **Biometric Privacy**: Streaming in-cabin facial video creates GDPR, CCPA and biometric-data compliance liabilities.
- **Hardware Cost**: Commercial vision systems rely on 3D-CNN or YOLO backbones that need accelerators drawing 30 W or more [6].
- **Motivation**: Build a private, sub-20 ms DMS that deploys on commodity laptop and embedded processors [3].

*Proprietary material of SILVER OAK UNIVERSITY

Relevance and Industrial Importance

- **Personal Calibration**: A 15-second resting baseline removes eye-shape bias and suppresses false alarms for narrow-eyed drivers [7].
- **12-Channel Kinematic Fusion**: Relative EAR, MAR, head pose, blink duration and the velocities ΔEAR, ΔMAR and ΔPitch feed continuous sliding windows [11].
- **Fleet Deployability**: Suited to trucking fleets, buses and ride-hailing services using off-the-shelf USB webcams.
- **No Sensor Retrofit**: Single-camera monitoring avoids infrared eye-trackers, EEG headbands and capacitive steering wheels.
- **Deterministic Edge Safety**: Sub-16 ms model latency supports immediate visual and audio alerts.

*Proprietary material of SILVER OAK UNIVERSITY

Literature Review: Part 1

Sr.	Reference	Architecture	Dataset	Result	Key Findings and Limitations
1	Liu et al. (2022) [1]	CNN-LSTM hybrid	Private driving video	94.2% accuracy	Validates temporal modeling over single-frame detection; lacks baseline calibration.
2	Sengupta et al. (2023) [2]	Systematic review	Survey (NTHU, UTA-RLDD)	Benchmark study	Confirms hybrid geometric-temporal models balance accuracy and latency best.
3	Farooq et al. (2024) [3]	VigilEye (landmark, CPU)	Custom webcam stream	<45 ms latency	Shows landmark pipelines on CPU are viable for real-time embedded DMS.
4	Ghoddoosian et al. (2021) [4]	UTA-RLDD baseline (HMM)	UTA-RLDD (60 subjects)	65.2% macro-F1	Established the naturalistic drowsiness benchmark; highlights subject variance.
5	Choi & Park (2023) [5]	MediaPipe + PERCLOS	Simulated video stream	93.8% accuracy	Finds MediaPipe Face Mesh 3× faster than dlib on commodity CPUs.

*Proprietary material of SILVER OAK UNIVERSITY

Literature Review: Part 2

Sr.	Reference	Architecture	Dataset	Result	Key Findings and Limitations
6	Al-Haija et al. (2023) [6]	YOLOv8-Nano DMS	Real-time webcam	91.5% mAP	Lightweight detector, but has no long-range temporal memory across frames.
7	Zhang et al. (2022) [7]	Adaptive EAR threshold	NTHU-DDD	96.1% accuracy	Justifies driver-specific calibration to avoid false alarms from narrow eyes.
8	Babu et al. (2021) [8]	Keypoints + BiLSTM	In-house video corpus	95.4% accuracy	Bidirectional recurrent units capture the onset of blink patterns.
9	Kumar et al. (2024) [9]	SAFE-DRIVE-AI (attention)	Multi-modal DMS	97.2% recall	Temporal self-attention weights critical micro-sleep frames highly.
10	Martinez et al. (2023) [10]	Head pose + landmark fusion	Driving simulator	94.8% accuracy	Fusing head pitch with eye closure detects nodding and fatigue.

*Proprietary material of SILVER OAK UNIVERSITY

Research Gap

- **Anatomical Baseline Disparity**: Across 48 subjects, resting open-eye EAR spans 0.18–0.45 (a 2.5× spread), so static thresholds label narrow-eyed alert drivers as drowsy.
- **Session-Level Labels**: UTA-RLDD labels cover 10-minute sessions, so open-eyed Low-Vigilant frames overlap geometrically with Alert frames.
- **Data-Leakage Inflation**: Reports above 90% accuracy on UTA-RLDD use random frame splits; strict subject-independent evaluation caps F1 at about 52–58%.
- **Central Attractor Collapse**: Symmetric cross-entropy treats boundary classes equally and pulls optimizers toward the middle class (86% LowVig recall).
- **Window Truncation**: 3–4 s windows cut off yawn cycles (3.5–5.5 s) and head-nod recoveries (2.0–4.5 s).

*Proprietary material of SILVER OAK UNIVERSITY

Objectives and Milestones Achieved

- **Real-Time Edge Inference**: Sequence models under 20 ms on a single CPU thread, no GPU. **Achieved: 15.30 ms (65.3 FPS).**
- **Personal Calibration**: A 15-second open-eye baseline maps any face to a canonical scale. **Achieved: stable in live webcam tests.**
- **Kinematic Velocity Pipeline**: 12 geometric channels capture eyelid, mouth and head velocity over 8.0 s windows. **Achieved: T = 120, F = 12.**
- **Ordinal Loss**: Alert-vs-Drowsy errors cost 4× more than boundary errors. **Achieved: $\lambda_{ord} = 0.35$.**
- **Complete System**: 9-model ensemble, debounced two-tier alarm and HUD. **Achieved: 57/57 integration tests pass.**

*Proprietary material of SILVER OAK UNIVERSITY

Dataset and Preprocessing

- **UTA-RLDD**: 60 subjects and 141 HD video recordings (about 2.54 million frames) with natural circadian fatigue progression [4].
- **Subject-Independent Folds**: 3-fold cross-validation with no subject overlap (Fold 1: subjects 01–15; Fold 2: 16–30; Fold 3: 31–45).
- **Two-Pass Extraction**: Pass 1 takes personal baselines (EAR_{base}, $Pitch_{base}$) from Alert videos only; Pass 2 extracts calibrated 120-frame sequences.
- **Faster Decoding**: Grabbing compressed packets with cap.grab() skips full decoding of dropped frames, giving an 8.9× extraction speedup.
- **Sequence Cache**: 26,050 calibrated sequences, tensor shape (26050, 120, 12), normalized the same way as live inference.

*Proprietary material of SILVER OAK UNIVERSITY

Required Tools and Technology

- **Deep Learning**: PyTorch 2.x, automatic mixed precision (fp16), Intel MKL and OpenMP CPU kernels.
- **Facial Geometry**: MediaPipe Tasks FaceLandmarker with the XNNPACK CPU delegate; 468 3D landmarks at about 15 ms [3], [5], [14].
- **Computer Vision**: OpenCV (VideoCapture, MJPG stream, solvePnP head pose with Levenberg–Marquardt optimization), NumPy, SciPy.
- **Audio and HUD**: Pygame procedural tone and siren synthesis with no external assets; OpenCV telemetry overlay.
- **Test Hardware**: Intel Core i7 / AMD Ryzen CPU (single thread), NVIDIA RTX 3050 Laptop GPU, 720p USB webcam, under 800 MB RAM.

Methodology: Kinematic Pipeline

- **Stage 1, Landmarks**: A 640×480 frame goes to MediaPipe Tasks, which returns 468 3D landmarks in about 15 ms.
- **Stage 2, Kinematic Differentials**: Velocity of each signal over $\delta t = 1/FPS$:

$$\Delta x(t) = [x(t) - x(t - \delta t)] / \delta t, \quad x \in \{EAR, MAR, Pitch\}$$

  - ΔEAR separates snap blinks from slow micro-sleep re-opening.
  - ΔMAR separates rapid speech from continuous yawn expansion.
  - $\Delta Pitch$ captures sudden downward head drops and startle jerks.
- **Stage 3, Calibrated PERCLOS (P_{60})**: Share of frames in the rolling buffer with $EAR(t) < 0.75 \times EAR_{base}$.
- **Stage 4, Biological Windows**: 120-frame sliding windows (8.0 s at 15 FPS) hold whole yawn cycles and nod recoveries.

*Proprietary material of SILVER OAK UNIVERSITY

Methodology: Calibration and Loss

- **Baseline Normalization**: Maps any face to a scale where 1.0 is open and ≤ 0.5 is closed eyes:

$$EAR_{rel}(t) = EAR_{avg}(t) / EAR_{base}, \quad Pitch_{rel}(t) = Pitch(t) - Pitch_{base}$$

- **Baseline Variance**: Measured baselines span 0.18–0.45 (2.5×), so a static threshold fails across subjects.
- **Ordinal Focal Loss**: Balanced focal loss [13] plus a quadratic ordinal penalty:

$$L_{total} = L_{Focal} + \lambda_{ord} \Sigma_{c=0..2} p_c (c - y_{true})^2$$

- **Asymmetric Penalty**: For an Alert sample (y = 0), predicting LowVig costs $1^2 = 1$ and predicting Drowsy costs $2^2 = 4$, a 4× stronger push; $\lambda_{ord} = 0.35$.

*Proprietary material of SILVER OAK UNIVERSITY

Architecture: Edge Model Triad

- **Triad Diversity**: Three models with different inductive biases remove single-architecture blind spots within an edge memory budget [12].
- **HybridCNNGRU** (18,131 parameters, 0.069 MB): 1D-CNN (12 to 32 channels, k = 3; 32 to 64, k = 5) filters local blink velocity; a GRU (hidden 48) with Bahdanau attention tracks fatigue onset.
- **TCNClassifier** (96,499 parameters, 0.368 MB): 5 causal dilated residual blocks (d = 1, 2, 4, 8, 16; k = 3) give a 125-frame receptive field for yawns and head droop [12].
- **BiGRUAttention** (37,971 parameters, 0.145 MB): Bidirectional GRU (hidden 36×2) with 4-head self-attention models both eye-closure onset and recovery.

*Proprietary material of SILVER OAK UNIVERSITY

Architecture: Ensemble and Alert Logic

- **Soft-Voting Fusion**: Averages class probabilities over M = 9 models to cancel single-model bias:

$$P_{avg}(c) = (1/M)\,\Sigma_{m=1..M}\,P_m(c \mid W), \quad c \in \{0, 1, 2\}$$

- **Online EMA**: $\alpha = 0.35$ (half-life about 106 ms) removes single-frame flicker:

$$P_{ema,t}(c) = \alpha\,P_{avg,t}(c) + (1 - \alpha)\,P_{ema,t-1}(c)$$

- **Two-Tier Debounced State Machine**:
  - P_{ema}(Alert) ≥ 0.50: **NORMAL** (green HUD, alarm counters reset).
  - P_{ema}(LowVig) ≥ 0.50 for ≥ 3.0 s (45 frames): **WARNING** (yellow HUD, single chime).
  - P_{ema}(Drowsy) ≥ 0.45 for ≥ 1.5 s (22 frames): **CRITICAL** (red HUD, repeating siren).

*Proprietary material of SILVER OAK UNIVERSITY

Results: Training Challenges

- **Phase 1, Central Attractor Collapse**: The first BiGRU labeled 86.0% of samples Low Vigilant (macro-F1 0.3428). Fixed with ordinal focal loss and a quadratic distance penalty.
- **Phase 2, Validation Divergence**: Validation loss rose from 0.61 to 0.74 because of camera-mount angle bias. Fixed with window pose median centering and AdamW weight decay (1×10^{-2}).
- **Phase 3, Train-Serve Skew**: Window-p90 normalization erased absolute eye closure, so alert live users read as Low Vigilant. Fixed with personal Alert-baseline scaling.
- **Window Length**: Extending windows to 120 frames (8.0 s) removed yawn-truncation errors.

*Proprietary material of SILVER OAK UNIVERSITY

Results: Holdout Performance

Model	Holdout fold	Macro-F1	Accuracy	F1 Alert	F1 LowVig	F1 Drowsy
CNN-GRU	Fold 1	0.5252	52.54%	0.4801	0.5076	0.5878
CNN-GRU	Fold 2	0.4945	49.89%	0.4397	0.4466	0.5973
CNN-GRU	Fold 3	**0.6062**	**60.37%**	**0.6622**	**0.5118**	**0.6445**
TCN	Fold 1	0.4986	49.51%	0.5502	0.3989	0.5468
TCN	Fold 2	0.4872	50.46%	0.4122	0.3870	**0.6623**
TCN	Fold 3	0.5430	54.52%	0.6407	0.4367	0.5516
BiGRU	Fold 1	0.5361	53.43%	0.5161	0.5132	0.5789
BiGRU	Fold 2	0.4741	47.67%	0.4486	0.4021	0.5715
BiGRU	Fold 3	0.5472	54.63%	0.5826	0.4311	0.6280
Ensemble average	**—**	**0.5236**	**52.56%**	**0.5258**	**0.4483**	**0.5965**

- **Ensemble mean**: 0.5236 macro-F1 and 52.56% accuracy on unseen subjects, within the 52–58% range expected for strict subject-independent evaluation.
- **Best single model**: CNN-GRU on Fold 3 reaches 0.6062 macro-F1 and 60.37% accuracy. Fold 2 is the hardest holdout for all three architectures.

Results: Latency and Edge Viability

Model	Parameters	Checkpoint	CPU latency (mean)	CPU throughput	GPU latency (RTX 3050)	Edge suitability
HybridCNNGRU	18,131	0.069 MB	15.30 ms	65.3 FPS	4.56 ms	Strict edge (<20 ms)
BiGRUAttention	37,971	0.145 MB	42.04 ms	23.8 FPS	2.91 ms	Soft real-time
TCNClassifier	96,499	0.368 MB	51.43 ms	19.4 FPS	12.36 ms	GPU recommended
Full 9-model ensemble	610,404	2.35 MB	511.65 ms	2.0 FPS	80.73 ms (12.4 FPS)	GPU real-time

- **Strict Edge Execution**: HybridCNNGRU needs 15.30 ms on a CPU (65.3 FPS), meeting the automotive real-time deadline without a GPU.
- **Compact Footprint**: The whole ensemble is 2.35 MB on disk, so the weights are small enough for any edge device.
- **Deployment Trade-off**: The full ensemble runs in real time only on a GPU (80.73 ms); on a CPU it manages 2.0 FPS.

*Proprietary material of SILVER OAK UNIVERSITY

System Verification

- **Integration Tests**: The suite covers normalizers, ensemble fusion and the alert manager; 57 of 57 tests pass in 2.75 s.
- **15-Second Calibration**: The live webcam routine stores the user's baseline (EAR_{base}, $Pitch_{base}$) in ~/.alert/calibration.json.
- **Awake Driver Test**: With eyes open naturally, P_{ema}(Alert) stays near 0.85–0.96 and the HUD remains green (NORMAL) with no false warnings.
- **Micro-Sleep Test**: Eyelid closure held past 1.5 s moves the system to CRITICAL (Drowsy) and sounds the siren within 22 frames.
- **HUD Telemetry**: Status pill, ensemble badge, ΔEAR and ΔPitch indicators, probability bars and an attention heatmap.

*Proprietary material of SILVER OAK UNIVERSITY

Conclusion

- **Edge DMS Realized**: An end-to-end, zero-cloud driver fatigue monitor, designed, trained and verified on edge-class hardware.
- **Anatomical Bias Removed**: Personal baseline normalization (EAR / EAR_{base}) removes the 0.18–0.45 cross-subject spread and lifted holdout F1 by 24.8%.
- **Kinematic Features Matter**: ΔEAR and ΔPitch separate voluntary snap blinks from involuntary micro-sleep closure.
- **Ordinal Formulation**: Penalizing opposite-extreme errors 4× harder prevents mode-seeking collapse and steadies boundary predictions.
- **Holdout Result**: Best model reaches 0.6062 macro-F1 and 60.37% accuracy on unseen subjects; the 9-model mean is 0.5236.

*Proprietary material of SILVER OAK UNIVERSITY

Future Enhancements

- **Near-Infrared Illumination**: Add 850 nm / 940 nm infrared camera streams for reliable night operation and polarized sunglasses.
- **Quantization and Acceleration**: Export to ONNX Runtime INT8, OpenVINO and TensorRT for Raspberry Pi 5, NXP BlueBox and Jetson Orin Nano.
- **CAN-Bus Fusion**: Combine visual alertness with steering-angle entropy, pedal pressure variance and lane-departure signals.
- **Driver Identification**: Use on-device face recognition to load each driver's calibration profile automatically, with no cloud.

*Proprietary material of SILVER OAK UNIVERSITY

References

[1] W. Liu, J. Qian, Z. Yao, X. Jiao, and J. Pan, “Real-time driver fatigue detection based on CNN-LSTM and facial landmark points,” IET Intelligent Transport Systems, vol. 16, no. 7, pp. 936–948, 2022.

[2] A. Sengupta, V. Patel, and R. Kumar, “Deep Learning Approaches for Driver Drowsiness and Distraction Detection: A Systematic Review,” Sensors, vol. 23, no. 4, art. 1952, 2023.

[3] M. A. Farooq, S. Corcoran, and P. Corcoran, “VigilEye: Lightweight Facial Landmark-Based Driver Drowsiness Detection Optimized for Real-Time Embedded CPU Inference,” arXiv:2403.08812, 2024.

[4] R. Ghoddoosian, M. Bartlett, and V. Athitsos, “A Realistic Dataset and Baseline Temporal Model for Driver Drowsiness Detection,” IEEE Access, vol. 9, pp. 110595–110607, 2021.

[5] H. K. Choi and J. H. Park, “Real-Time Driver Fatigue Monitoring System Using Google MediaPipe Face Mesh and Temporal PERCLOS Analysis,” Applied Sciences, vol. 13, no. 11, art. 6672, 2023.

[6] Z. Al-Haija, M. Smadi, and M. Al-Zoubi, “Real-Time Driver Monitoring System Using Lightweight YOLOv8-Nano and Edge Computing Architectures,” Eng. Appl. Artif. Intell., vol. 126, art. 106981, 2023.

[7] Y. Zhang, H. Huang, and Y. Wang, “Personalized Driver Fatigue Assessment via Adaptive Eye Aspect Ratio Thresholding and Blink Dynamics,” Pattern Recognition Letters, vol. 158, pp. 124–131, 2022.

[8] K. S. Babu, S. R. Chalamala, and K. S. Rao, “Driver Drowsiness Classification Using Facial Keypoint Vectors and Bidirectional LSTM Networks,” IEEE Trans. Intell. Transp. Syst., vol. 22, no. 10, pp. 6423–6434, 2021.

References

[9] D. Kumar, R. Sharma, and A. Gupta, “SAFE-DRIVE-AI: Temporal Self-Attention BiGRU for Micro-Sleep and Low-Vigilance Detection,” Expert Systems with Applications, vol. 238, art. 122115, 2024.

[10] J. B. Martinez, L. Gomez, and F. Rossi, “Multimodal Fusion of Head Pose Pitch and Facial Geometric Indicators for Non-Intrusive Driver Drowsiness Assessment,” Accident Analysis & Prevention, vol. 182, art. 106945, 2023.

[11] T. Soukupova and J. Cech, “Real-time eye blink detection using facial landmarks,” 21st Computer Vision Winter Workshop (CVWW), Rimske Toplice, Slovenia, 2016, pp. 1–8.

[12] S. Bai, J. Z. Kolter, and V. Koltun, “An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling,” arXiv:1803.01271, 2018.

[13] T. Lin, P. Goyal, R. Girshick, K. He, and P. Dollar, “Focal Loss for Dense Object Detection,” IEEE Trans. Pattern Anal. Mach. Intell., vol. 42, no. 2, pp. 318–327, 2020.

[14] C. Lugaresi et al., “MediaPipe: A Framework for Building Perception Pipelines,” arXiv:1906.08172, 2019.

[15] European Commission, “Regulation (EU) 2019/2144 of the European Parliament and of the Council on type-approval requirements for motor vehicles (General Safety Regulation),” Official Journal of the European Union, L 325, pp. 1–40, 2019.